

MCV HW4 — Theory

Modern Computer Vision — Technion, Spring 2026

Due: July 22, 2026

Submission: a single PDF (typed or a clear scan of handwritten work). Show your reasoning — a bare final answer earns no credit. Each question is worth **50%** of the theory grade.

1. Suppose we are given two homographies that were extracted from points on two distinct planes. Unfortunately, the correspondences the homographies were extracted with were not saved. Can we still extract the fundamental matrix? Show how.

Hint: Show that $H^T F$ is skew-symmetric.

Recall: M is skew-symmetric $\iff x^T M x = 0 \ \forall x \iff M + M^T = 0$.

2. How many fixed points $\left(p = \frac{H(p)}{H(p)_z}\right)$ can a non-trivial ($H \neq \alpha I$) homography have? How many lines can be *pointwise fixed* (i.e. every point on the line is fixed)?